Worksheet 17

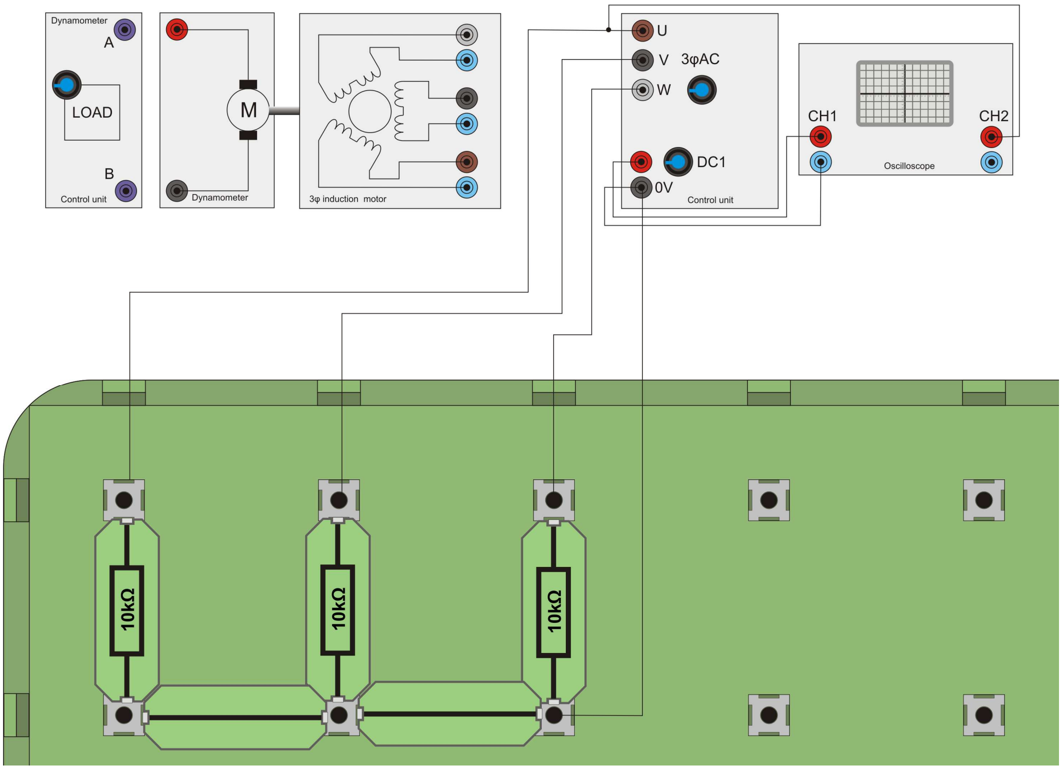
Three-phase motor drive strategies

Modern electrical machines

The National Grid distributes electricity in three separate phases rather than the single-phase found in homes. Three-phase generation is more energy efficient.

Factories use three-phase motors as they too are more efficient than single-phase systems.

The photograph shows a three-phase induction motor factory in China.



Over to you:

1) Set up the three-phase induction motor as shown in the diagram above. As this exercise looks only at the signals the generator unit creates, there is no need to connect the three-phase motor to the supply.
2) Connect three 10kΩ resistors as loads between U, V, W and 0V.
3) On your PC run 'Open Control_3Phase.bat'.
4) Press 'Play' and put the software into digital mode by pressing the 'Sine' button. This changes the waveform from pseudo-sine, constructed with PWM to a simple digital waveform where U, V, and W are 120 degree shifted square waves. Set the output frequency to 50Hz and press 'Run'.
5) Trigger an oscilloscope on the DC1 output. On the second channel display the output U.
6) Sketch the waveforms of signals U, V, and W , relative to the timing waveform on DC1.

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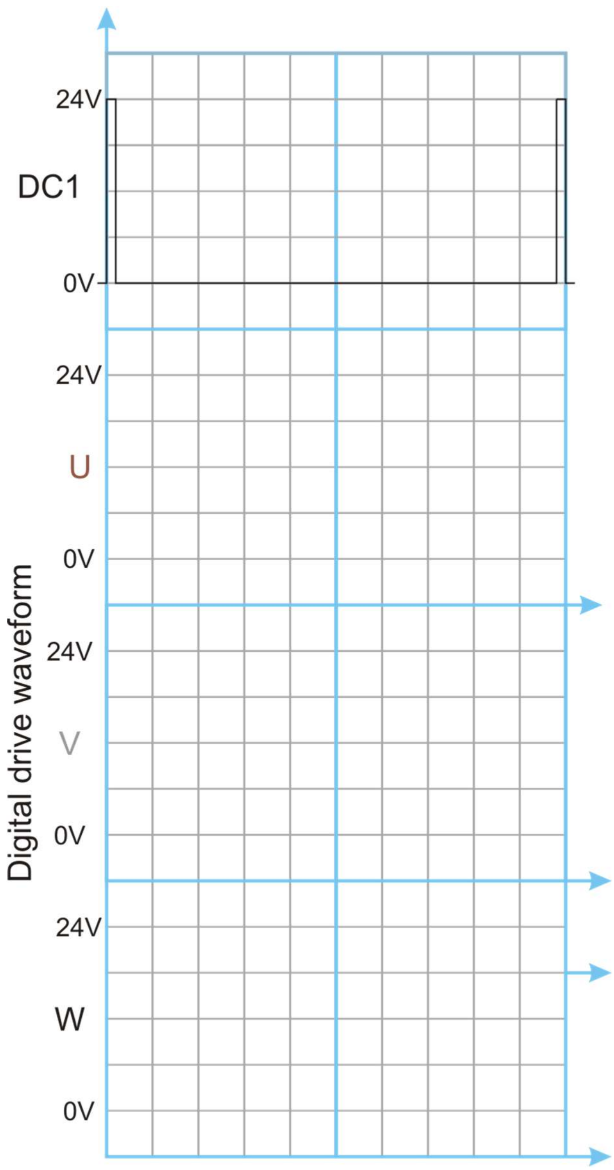

So what?

Notice that these pulses are 120^{o} apart. Together, they make up the basic three-phase waveform.

You should notice that each waveform spends some time at 0V, some time at 24V and some time when it is in between, at around a volt or so. What you are seeing is the three stages of drive of each output - forced to 0V, forced to +24V and left open circuit. In this last stage, the voltage is not dictated by the power supply but by the rest of the circuit.

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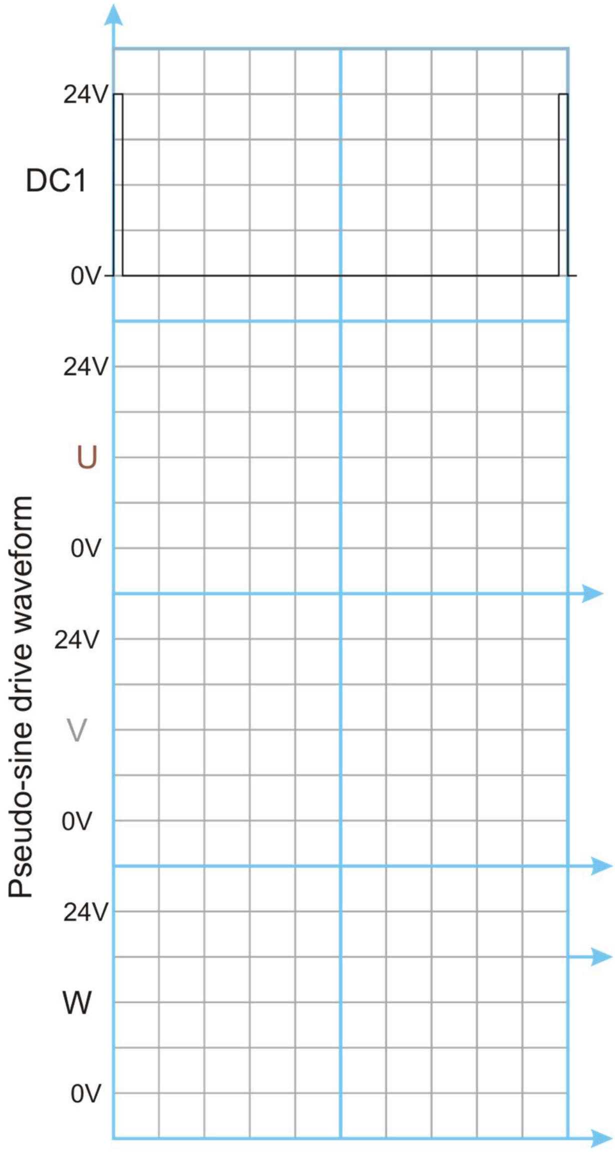
Over to you:

1) Still with the 10kΩ resistive loads in place, change the drive waveform to pseudo sine at 50Hz.

4) Once again, sketch the waveforms of signals U, V, and W , relative to the timing waveform on DC1.



So what?

You now see the waveforms that make up the three-phase drive. There are in fact six separate steps to this. Hence it is often called the 'six step three-phase motor drive system'.

Some three-phase speed control systems use simple square waves for driving motors. However the pseudo-sine wave you have examines is much more efficient.